Proof of M.C.4.4

Course IV – Risk Management

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Proposition:

$$VAR(P/L_{fix}) < VAR(P/L_{Float}) \Leftrightarrow \rho_{EBIT,I_{Float}} < \frac{\sigma_{I_{Float}}}{2\sigma_{EBIT}}$$

PART I

If $VAR(P/L_{fix}) < VAR(P/L_{Float})$, *then* $\rho_{EBIT,I_{Float}} < \frac{\sigma_{I_{Float}}}{2\sigma_{EBIT}}$

Proof (Direct): Suppose it holds

$$VAR(P/L_{fix}) < VAR(P/L_{Float})$$

From the unit materials, we know that

$$VAR(P/L_{fix}) = VAR(EBIT) \text{ and}$$
$$VAR(P/L_{Float}) = VAR(EBIT) + VAR(I_{I_{Float}}) - 2Cov(EBIT, I_{Float})$$

Substituting both sides, we have

$$VAR(EBIT) < VAR(EBIT) + VAR(I_{Float}) - 2Cov(EBIT, I_{Float}).$$

Now, observe the correlation formula

$$\rho_{X,Y} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

Rearranging for Cov(X,Y) and plugging it into the inequality we obtain

$$VAR(EBIT) < VAR(EBIT) + VAR(I_{Float}) - 2\rho_{EBIT,I_{Float}} \sigma_{EBIT} \sigma_{I_{Float}}$$

Let's now rearrange the inequality

$$VAR(EBIT) - VAR(EBIT) < VAR(I_{Float}) - 2\rho_{EBIT,I_{Float}} \sigma_{EBIT} \sigma_{I_{Float}}$$

$$2\rho_{EBIT,I_{Float}} \sigma_{EBIT} \sigma_{I_{Float}} < VAR(I_{Float})$$

$$\rho_{EBIT,I_{Float}} < \frac{VAR(I_{Float})}{2\sigma_{EBIT} \sigma_{I_{Float}}}$$

Using the definition of variance $VAR(X) = \sigma_X^2$ we simplify it to the following result

$$\rho_{EBIT,I_{Float}} < \frac{\sigma_{I_{Float}}^2}{2\sigma_{EBIT} \sigma_{I_{Float}}}$$

$$\rho_{EBIT,I_{Float}} < \frac{\sigma_{I_{Float}}}{2\sigma_{EBIT}}$$

Essentially, this result allows us to estimate the boundary value for volatility of the floating rate $\sigma_{I_{Float}}$, by which we decide whether to choose fixed rate instrument or floating rate instrument in order to minimize the volatility of pre-tax P/L ratio. If $\sigma_{I_{Float}}$ is greater than this value, we choose fixed rate instrument

PART II

If $\rho_{EBIT,I_{Float}} < \frac{\sigma_{I_{Float}}}{2\sigma_{EBIT}}$, then $VAR(P/L_{fix}) < VAR(P/L_{Float})$

Proof (Direct): Conversely, suppose it holds that

$$\rho_{EBIT,I_{Float}} < \frac{\sigma_{I_{Float}}}{2\sigma_{EBIT}}$$

By the following rearrangements and substitutions using previous results we get

$$\rho_{EBIT,I_{Float}} < \frac{VAR(I_{Float})}{2\sigma_{EBIT}\sigma_{I_{Float}}}$$

$$2\rho_{EBIT,I_{Float}}\sigma_{EBIT}\sigma_{I_{Float}} < VAR(I_{Float})$$

$$VAR(EBIT) - VAR(EBIT) < VAR(I_{Float}) - 2\rho_{EBIT,I_{Float}}\sigma_{EBIT}\sigma_{I_{Float}}$$

$$VAR(EBIT) < VAR(EBIT) + VAR(I_{Float}) - 2\rho_{EBIT,I_{Float}}\sigma_{EBIT}\sigma_{I_{Float}}$$

$$VAR(EBIT) < VAR(EBIT) + VAR(I_{Float}) - 2Cov(EBIT, I_{Float})$$

$$VAR(P/L_{fix}) < VAR(P/L_{Float})$$

$\square$